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Research Statement

Research interests: Applied Microeconomics, Production Econometrics, and Industrial Organization.

I study how technology adoption changes firm productivity and market power, and why its benefits differ across firms. I combine economic models with firm-level data to distinguish productive efficiency from pricing behavior. I have publications in *Economic Modelling* and *Economics Letters* on Nigerian banking productivity and heterogeneous treatment-effect methods. My job market paper studies Spanish manufacturing and is under review. In 2025, I received the Best Third-Year Paper Award in Binghamton University's Department of Economics.

My independent agenda develops two connected directions: recovering reliable economic quantities from flexible production models, and understanding how technology's benefits spread beyond adopting firms. My solo regularization project asks whether accurate output prediction also yields reliable elasticities and markups. My planned solo AI project examines productivity spillovers through geographic proximity and economic linkages. These projects extend my dissertation toward questions about measurement and the distribution of technology's gains.

Job Market Paper: Digitalization, Productivity, and Market Power

Higher revenues do not reveal how digital investments change production. Technology may improve the efficiency of labor or of the whole input bundle. It may also change prices relative to costs. My job market paper, **"Digitalization, Factor-Biased Productivity, and Market Power"** (with Subal C. Kumbhakar), distinguishes these channels within a common framework. The application uses the Encuesta Sobre Estrategias Empresariales (ESEE), a panel of Spanish manufacturing firms over 2000–2019.

The digitalization index combines thirteen indicators covering IT, digital human capital, automation, and interactions with customers and suppliers. Technology is identified under imperfect competition using dynamic production moments. Revenue expenditure shares then provide a markup signal; lagged instruments identify its dynamics. Identification does not rely on a price-based materials-demand inversion. Additional state-space restrictions separate persistent components from output noise in firm-year predictions.

On a common estimation sample, a 0.10 increase in the digitalization index is associated with next-period labor-augmenting and Hicks-neutral productivity responses of approximately 0.48 and 0.65 percent under perfect competition. Under imperfect competition, the corresponding estimates are **0.55 and 0.43 percent**, and the markup response is 0.07 percent. These are conditional relationships under the model's assumptions.

Allowing for market power significantly shifts the relative productivity response toward labor efficiency. The point estimates reverse order, but neither within-model difference is statistically significant on the common sample. Hicks-neutral productivity still contributes more to output because labor-augmenting gains enter through the labor elasticity.

The labor-efficiency response is significantly larger among small firms; the large-firm estimate is imprecise. Markup responses are significantly smaller in digitally intensive industries. These differences motivate my work on adoption and the distribution of technology's gains.

Published Research: Technology, Performance, and Empirical Methods

My banking publication provides a complementary perspective on technology and firm performance. **"The Scale–Technology Paradox: Technological Progress and Productivity Decline in Nigerian Banking"** (with Subal C. Kumbhakar) is published in *Economic Modelling* (2026). It studies seventeen Nigerian commercial banks over 2000–2023. The analysis accounts for multiple outputs and endogenous input choices. It distinguishes changes in production possibilities from changes in realized productivity.

Estimated productivity declines by an average of **3.5 percent annually**, even though technical change reduces costs. Returns to scale average 1.46, indicating potential gains from expansion. Larger and internationally licensed banks experience steeper declines. The economic lesson is that modernization and greater scale need not improve realized performance. The results motivate attention to lending efficiency and the conditions that allow banks to use improved technologies effectively. The paper's public replication code makes the analysis available for further research.

My other publication, **"Causal Forests versus Penalized Splines for Heterogeneous Treatment Effects"** (with Ivan Korolev and Nency Dhameja), appears in *Economics Letters* (2026). It examines how the form of treatment-effect variation should guide empirical method choice. In simulations with randomized assignment, forests perform best when effects change sharply. Penalized splines generally perform better with smooth heterogeneity and in designs combining continuous, binary, and group-level covariates.

This paper informs my independent agenda by showing why method choice should reflect the quantity of interest and the structure of the problem. In the regularization project, that quantity is an elasticity rather than a predicted output. In future adoption research, heterogeneous-effect methods can help examine which firms benefit and which complementary investments explain those differences.

Working Paper: The Channels of Digitalization

In **"Mediation Analysis in Production Functions with Unobserved Productivity"** (with Subal C. Kumbhakar), I study how digitalization's relationship with output operates through changes in labor hours. More productive firms may both adopt technology and adjust employment. This makes it difficult to distinguish a labor-mediated effect from differences in underlying productivity.

The paper develops a framework that controls for unobserved productivity in both the production and labor-hours equations. Simulations show that omitting this control from the labor equation can substantially overstate the indirect effect. An application to Spanish manufacturing distinguishes producing more through additional labor from producing more efficiently. This distinction supports my broader effort to identify what technology changes within firms.

Independent Agenda: Reliable Productivity Measurement

In my solo project, **"Regularization in Translog Production Models,"** I ask whether better output prediction improves the economic conclusions drawn from production models. Flexible models allow an input's contribution to vary with a firm's input mix. Their squared and interaction terms can, however, be highly correlated. A model may therefore predict output accurately while producing unstable input elasticities. This matters because elasticities underpin estimates of returns to scale and production-based markups.

I am developing a comparison of penalized regression and dimension-reduction methods. The planned simulations will vary input correlations, sample size, and the underlying production relationships. I will compare predictive accuracy with the bias and variability of recovered elasticities. I will also assess returns to scale and the frequency of economically implausible estimates, such as negative marginal products. A further step will trace how elasticity errors affect inferred markups.

The contribution I seek is guidance on selecting methods for economic interpretation. A method chosen to minimize prediction error may perform differently when the goal is to recover marginal products or market power. I plan to assess regularization alongside approaches that address endogenous input choices. Regularization alone does not resolve that problem. This project will establish when greater numerical stability supports reliable economic conclusions and when it conceals distortions.

Independent Agenda: AI and Productivity Spillovers

My planned solo project, **"Artificial Intelligence and Spatial Factor-Biased Productivity,"** asks who benefits from AI beyond the adopting firm. AI may improve the efficiency of particular inputs within firms. Knowledge flows, worker mobility, and supply-chain relationships may transmit gains to other firms. Geographic proximity and economic linkages may therefore expose firms to different forms of spillovers.

The first task is to identify an empirical setting with credible measures of AI use, production inputs and outputs, and exposure to other firms' adoption. I will assess whether the available data can distinguish geographic exposure from economic linkages. This will determine which channels the project can study. The design must also separate spillovers from common shocks and account for firms' selection into adoption. I will evaluate these requirements before interpreting exposure relationships as causal effects.

I plan to extend the adoption framework to distinguish spillovers that improve labor efficiency from gains affecting the whole input bundle. The project will also examine whether benefits vary with firms' own adoption and complementary investments. These questions connect technology diffusion to the uneven distribution of productivity gains. The project remains at the design stage; its empirical scope will depend on the data and identification opportunities.

Research Direction

My research program links measurement, mechanisms, and the distribution of technology's benefits. The dissertation establishes distinctions between productive efficiency, pricing, and input adjustment. My independent projects build on these distinctions to improve productivity measurement and examine gains beyond adopting firms. Across these projects, I will evaluate empirical methods by the economic conclusions they can support.